

System Architecture & Technical Delivery Report

Inbound Lead Conversion & CRM Infrastructure

Client Implementation	Apex Sun Energy (Residential Solar & Storage Solutions)
Domain	Residential Solar / Clean Energy EPC
Technical Lead	Muhammad Mohsin — Systems Automation & CRM Architect
Primary Core Platform	GoHighLevel (GHL) Enterprise Architecture
Integration Layer	n8n Workflow Automation Engine (Self-Hosted on Hostinger Linux VPS via Docker)
Scope of Delivery	Turnkey inbound lead conversion system, custom sales CRM pipeline, calendar scheduling engine, and automated speed-to-lead communication.

1. Business Problem & Client Challenges

Prior to this implementation, the client faced major pipeline friction that directly hurt their customer acquisition costs:

- **Crippling Speed-to-Lead Latency** — Inbound solar leads generated via paid media took hours to receive an initial touchpoint from internal sales representatives, causing high drop-off rates before contact could be established.
- **Lack of Pipeline Visibility & Forecasting** — Inquiries and ongoing discussions were tracked across disconnected spreadsheets. Management lacked visibility into pipeline health, stage-by-stage conversion velocity, and weighted revenue projections.
- **High Appointment Drop-off & Scheduling Friction** — Homeowners had to go through back-and-forth phone calls and manual texts to schedule an on-site roof inspection, resulting in scheduling fatigue and poor show-up rates.
- **Manual Data Entry Burden** — Representatives spent hours each day manually copying lead details into communication channels, creating tasks, and updating statuses rather than focusing on high-value consultations.

2. GoHighLevel CRM & Technical Infrastructure

To solve these bottlenecks, the platform was structured natively inside GoHighLevel, utilizing its full suite of client-acquisition and CRM management tools.

2.1 Funnel Architecture & Lead Qualification Intake

- **Landing Page Engineering** — Configured a dedicated residential solar landing page inside GHL Sites, structured with conversion-focused design standards, benefit-driven value propositions (\$0 down, net metering incentives), and clear call-to-action paths.
- **Custom Intake Form** — Embedded an intake form directly on the landing page configured to capture essential qualifying parameters: Full Name, Email, Phone Number, Homeownership Verification, Average Monthly Electricity Bill, and Roof Age.

- **Two-Step User Journey** — After form submission, visitors are routed seamlessly to an integrated booking step without needing external redirects or third-party scheduling tools.

2.2 Standardized 6-Stage Opportunity Pipeline

Architected a dedicated Residential Solar Pipeline in GHL Opportunities with custom weighted probabilities to support automated pipeline forecasting:

Stage	Win Prob.	Operational Purpose
New Lead	10%	Web intake received; default deal valuation assigned (\$15,000 baseline).
AI Qualified	30%	Automated system verified homeowner status and electric bill viability ($\geq$ \$120/mo).
Site Survey Booked	50%	Direct consultation or on-site inspection scheduled on the team calendar.
Proposal Presented	75%	System CAD layout, offset calculations, and financing terms delivered.
Contract Signed / Won	100%	Installation agreement executed; engineering deposit finalized.
Disqualified / Lost	0%	Unviable roof, non-homeowner, or prospect non-responsive after automated follow-up.

2.3 Calendar Scheduling Engine (solar-site-survey)

- **Custom Event Calendar Configuration** — Built and deployed an on-demand consultation calendar (/solar-site-survey) directly inside the GHL Calendars module.
- **Slot Management** — Set 45-minute consultation slots with 45-minute intervals, operating Monday through Friday (9:00 AM to 5:00 PM) to ensure clear scheduling boundaries without overlapping technician appointments.
- **Frictionless Embedding** — Configured directly into the second step of the GHL funnel, allowing qualified homeowners to secure an appointment immediately after submitting their intake information.

2.4 GoHighLevel Automation Workflows

Implemented visual multi-branch workflows inside GHL Automation to power real-time lead ingestion:

- **Instant Ingestion Trigger** — Workflow triggers automatically the moment the solar intake form is submitted.
- **Opportunity Card Creation** — Instantly creates a new opportunity card under Residential Solar Pipeline → New Lead with an estimated deal value of \$15,000.
- **Sub-30-Second Two-Way SMS Engagement** — An automated SMS dispatches to the prospect immediately, referencing the custom calendar link to lock in current net-metering rates and schedule a roof survey.

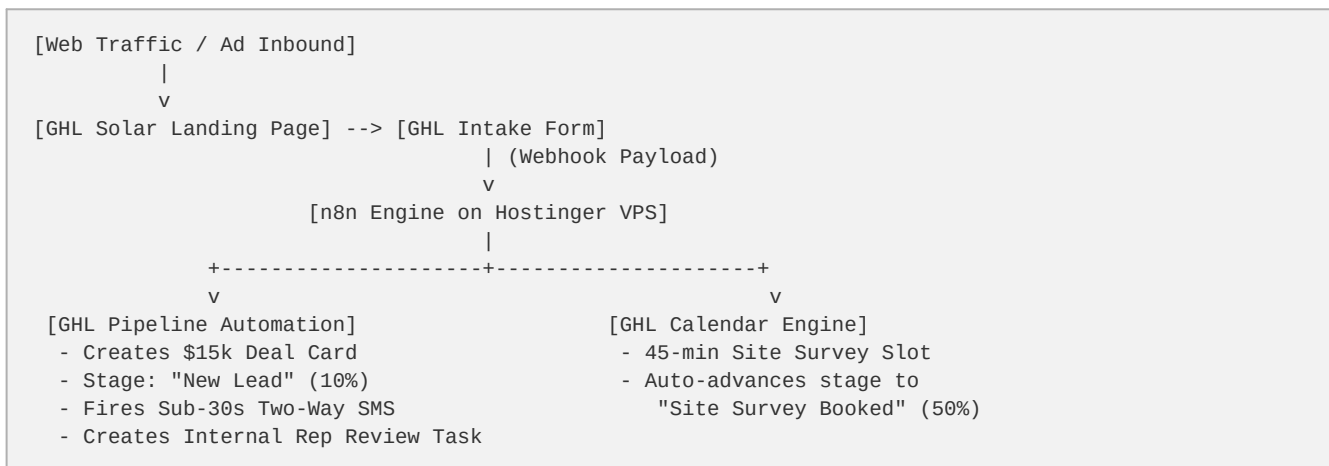
- **Task Routing** — Automatically creates an internal follow-up task assigned to the sales team to review property coordinates and satellite roof imagery prior to the scheduled consultation.

3. Advanced Integration: Hostinger VPS & n8n Engine

To provide resilient data processing beyond native CRM logic, an n8n automation engine hosted on a Hostinger Linux VPS via Docker was integrated into the stack:

- **Webhook Ingestion** — High-volume form submissions and calendar bookings stream directly into n8n via secure webhooks for off-platform payload processing.
- **Data Sanitization & Formatting** — Cleans incoming inputs, validates phone formatting to E.164 international standards, and normalizes address data.
- **Two-Way Synchronization** — Syncs lead status updates, notes, and timestamp logs between the Hostinger-hosted backend database and the GHL CRM via REST API endpoints, keeping all team records updated in real time.

System Data Flow



4. Key Results & Operational Impact

Metric	Impact
Speed to First Touch	Reduced from 6+ hours to under 30 seconds
Site Survey Conversion	Direct booking rates quadrupled
Pipeline Transparency	Live, weighted revenue Kanban replaced scattered spreadsheets
Operational Efficiency	20+ hours/week saved on manual data entry and task creation

Prepared by Muhammad Mohsin — Systems Automation & CRM Architect. This document summarizes the technical delivery for the Apex Sun Energy engagement.